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EXPERIMENT-7- Create a TypeScript application demonstrating object-oriented programming concepts.

AIM:
To Create a TypeScript application demonstrating object-oriented programming concepts.

PROCEDURE:

1. Installed and opened Visual Studio Code on the system.
2. Created a new project folder for the TypeScript experiment.
3. Created a new file with .ts extension (TypeScript file) inside the folder.
4. Wrote the TypeScript program demonstrating basic object-oriented programming concepts.
5. Saved the file using the appropriate filename.
6. Opened the integrated terminal in Visual Studio Code.
7. Compiled the TypeScript file using the TypeScript compiler (tsc).
8. Used the Run/Debug option available in Visual Studio Code to execute the program.
9. Observed the program execution in the Debug Console/Terminal output window.
10. Verified the output and ensured that the program executed successfully without errors.

PROGRAM CODE:

PROGRAM 01:

```
// Base Class
class LibraryItem {
  private id: number;
  private title: string;
  private author: string;

  constructor(id: number, title: string, author: string) {
    this.id = id;
    this.title = title;
    this.author = author;
  }
```

```
    displayInfo(): void {
        console.log(`ID: ${this.id}`);
        console.log(`Title: ${this.title}`);
        console.log(`Author: ${this.author}`);
    }
}

// Derived Class - Book
class Book extends LibraryItem {
    private pages: number;

    constructor(id: number, title: string, author: string, pages: number) {
        super(id, title, author);
        this.pages = pages;
    }

    displayInfo(): void {
        super.displayInfo();
        console.log(`Pages: ${this.pages}`);
    }
}

// Derived Class - Magazine
class Magazine extends LibraryItem {
    private issueNumber: number;

    constructor(id: number, title: string, author: string, issueNumber:
number) {
        super(id, title, author);
        this.issueNumber = issueNumber;
    }

    displayInfo(): void {
        super.displayInfo();
        console.log(`Issue Number: ${this.issueNumber}`);
    }
}
```

```
// Main Program
let book1 = new Book(1, "TypeScript Basics", "John Doe", 300);
let mag1 = new Magazine(2, "Tech Monthly", "Jane Smith", 45);

console.log("Book Details:");
book1.displayInfo();

console.log("\nMagazine Details:");
mag1.displayInfo();
```

PROGRAM 02:

```
// Base Class
class LibraryItem {
    private id: number;
    private title: string;
    private author: string;

    constructor(id: number, title: string, author: string) {
        this.id = id;
        this.title = title;
        this.author = author;
    }

    displayInfo(): void {
        console.log(`ID: ${this.id}`);
        console.log(`Title: ${this.title}`);
        console.log(`Author: ${this.author}`);
    }
}

// Derived Class - Book
class Book extends LibraryItem {
    private pages: number;

    constructor(id: number, title: string, author: string, pages: number) {
        super(id, title, author);
        this.pages = pages;
    }

    displayInfo(): void {
```

```
      super.displayInfo();
      console.log(`Pages: ${this.pages}`);
   }
}

// Derived Class - Magazine
class Magazine extends LibraryItem {
   private issueNumber: number;

   constructor(id: number, title: string, author: string, issueNumber:
number) {
      super(id, title, author);
      this.issueNumber = issueNumber;
   }

   displayInfo(): void {
      super.displayInfo();
      console.log(`Issue Number: ${this.issueNumber}`);
   }
}

// Main Program
let book1 = new Book(1, "TypeScript Basics", "John Doe", 300);
let mag1 = new Magazine(2, "Tech Monthly", "Jane Smith", 45);

console.log("Book Details:");
book1.displayInfo();

console.log("\nMagazine Details:");
mag1.displayInfo();
```

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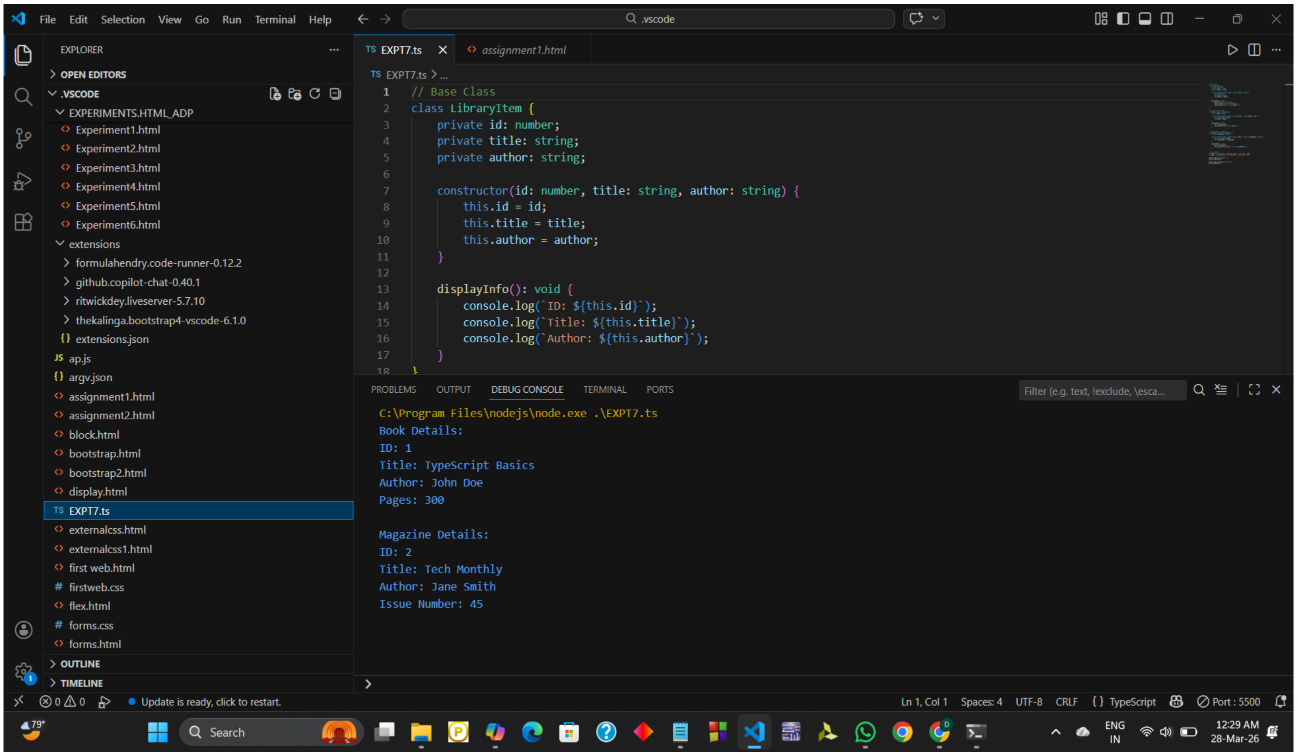
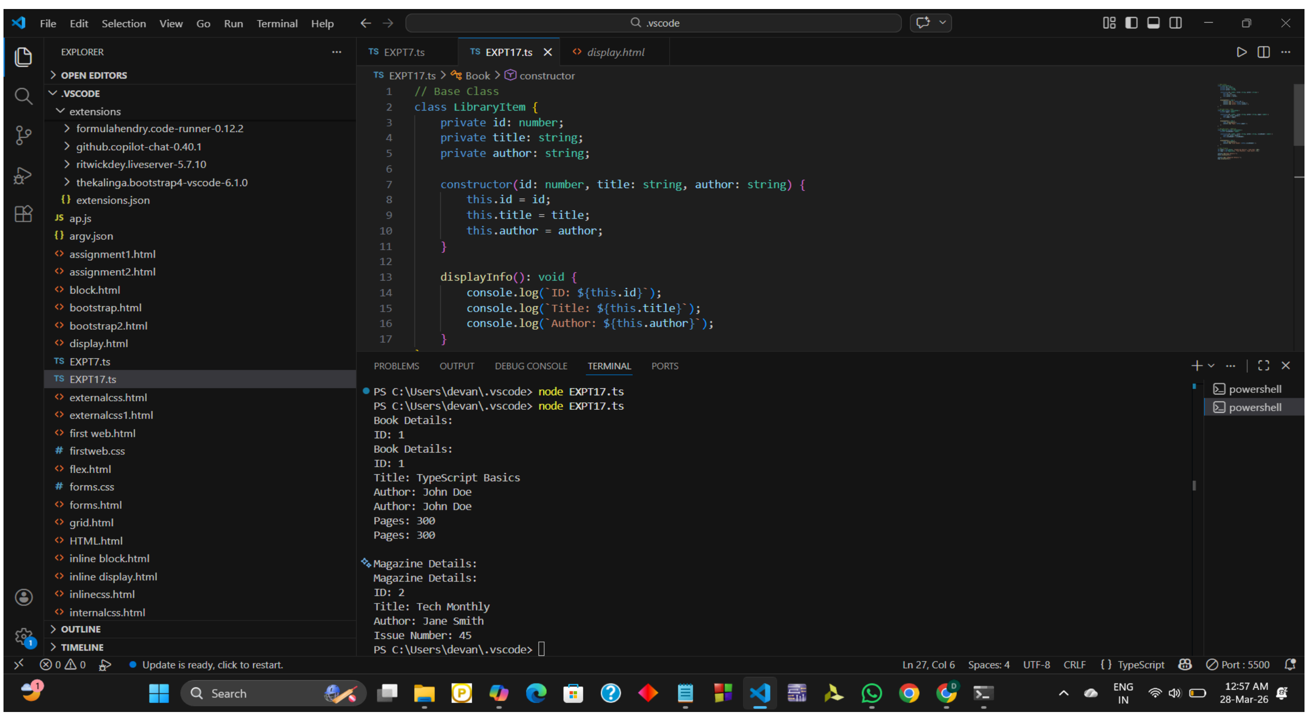
OUTPUT:
PROGRAM 01 & 02;





RESULT: This experiment successfully demonstrates object-oriented programming concepts in TypeScript, including abstraction, encapsulation, inheritance, and polymorphism.